

Lezione 3 - Studi di funzione

Matematica del continuo

Modulo 11 - Studio di funzioni

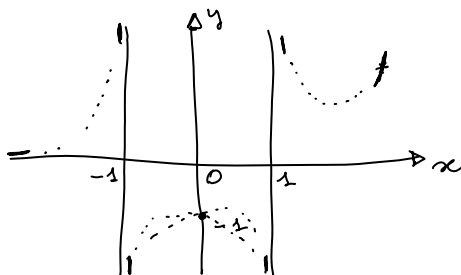
Unità didattica 2 – Funzioni con logaritmi ed esponenziali

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$$y = \frac{e^{2x}}{x^2 - 1} \quad \text{CE } (-\infty, -1) \cup (-1, 1) \cup (1, +\infty)$$

$y(0) = -1$
 linee $x \rightarrow -\infty \quad y = 0^+$ $\frac{e^{-\infty}}{+\infty} \quad \frac{0^+}{+\infty}$ asse x
 $\frac{0^+}{+\infty}$ AS ORIZZ
 per $x \rightarrow -\infty$
 linee $x \rightarrow +\infty \quad y = +\infty$ $\frac{+\infty}{+\infty}$ $\frac{+\infty}{+\infty}$ $\frac{+\infty}{+\infty}$ $\frac{+\infty}{+\infty}$
 $\frac{+\infty}{+\infty}$ $\frac{+\infty}{+\infty}$ $\frac{+\infty}{+\infty}$ $\frac{+\infty}{+\infty}$
 linee $x \rightarrow 1^+$ $y = \pm \infty$ $\frac{e^2}{1^+-1} \quad \frac{e^2}{0^+} / \frac{e^2}{1^--1} \quad \frac{e^2}{0^-}$
 $\frac{e^2}{1^+-1} \quad \frac{e^2}{0^+} / \frac{e^2}{1^--1} \quad \frac{e^2}{0^-}$
 linee $x \rightarrow -1^-$ $y = \pm \infty$ $\frac{e^2}{-1^+-1} \quad \frac{e^2}{0^+} / \frac{e^2}{-1^--1} \quad \frac{e^2}{0^-}$
 $\frac{e^2}{-1^+-1} \quad \frac{e^2}{0^+} / \frac{e^2}{-1^--1} \quad \frac{e^2}{0^-}$
 rette $x=1$
 AS. VERT
 Sx/dx
 rette $x=-1$
 AS. VERT
 Sx/dx



$$y = \frac{e^{2x}}{x^2 - 1}$$

$$y' = \frac{2e^{2x}(x^2 - 1) - e^{2x} \cdot 2x}{(x^2 - 1)^2}$$

$$= \frac{2e^{2x}(x^2 - x - 1)}{(x^2 - 1)^2}$$

$$y' \geq 0 \quad x^2 - x - 1 \geq 0$$

$$y' \geq 0$$

$$x^2 - x - 1 \geq 0$$

$$x = \frac{1 \pm \sqrt{1+4}}{2} < \frac{1+\sqrt{5}}{2}$$

$$x = \frac{1 - \sqrt{5}}{2}$$

